

HW2 for Modern Probability

Total Points: 100

In this problem set, we assume that all random variables are defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, unless specified otherwise.

1. Let $X_1, X_2, \dots$ be $\mathbb{R}$ -valued random variables and $a \in \mathbb{R}$. We define

$$\tau(\omega) = \inf\{n \geq 1 : X_n(\omega) > a\},$$

where the convention is $\inf \emptyset = \infty$.

- (a) [5 points] Show that τ is a random variable taking value in $\mathbb{N} \cup \{\infty\}$.
(b) [5 points] Define

$$Y(\omega) = \begin{cases} X_{\tau(\omega)}(\omega), & \tau(\omega) < \infty, \\ 0, & \tau(\omega) = \infty. \end{cases}$$

Show that Y is a $\mathbb{R}$ -valued random variable.

Hint: $Y(\omega) = \sum_{n=1}^{\infty} X_n(\omega) \mathbb{1}_{\{\tau(\omega) \geq n\}}$.

- (c) [5 points] Suppose that $X_n(\omega) \rightarrow X(\omega)$ for every ω . Show that $X(\omega) \leq a$ for $\omega \in \{\tau = \infty\}$.

[Total for Question 1: 15 points]

2. For a random variable X , recall that $\sigma(X) = \{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$ is the *smallest* σ -algebra with respect to which X is measurable.
- (a) [5 points] Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Borel function. Show that $\sigma(f(X)) \subset \sigma(X)$.
(b) [5 points] For $f(x) = e^x$, show that $\sigma(f(X)) = \sigma(X)$.
(c) [5 points] Let $\Omega = \{-1, 1\}$ and $X(\omega) = \omega$. Show that $\sigma(|X|) \subsetneq \sigma(X)$.
(d) [5 points] Let X and Y be independent random variables and f, g be Borel functions. Show that $f(X)$ and $g(Y)$ are independent.

[Total for Question 2: 20 points]

3. [10 points] Let $X_n, X \geq 0$. Suppose that $X_n \rightarrow X$ and $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$. Show that

$$\mathbb{E}|X_n - X| \rightarrow 0, \quad n \rightarrow \infty.$$

Hint: $|X_n - X| = X_n + X - 2\min(X_n, X)$ and use DCT on $\min(X_n, X)$.

[Total for Question 3: 10 points]

4. Let X be a non-negative random variable and $X_n = X \wedge n$.

- (a) [5 points] Show that $\mathbb{E}X_n \uparrow \mathbb{E}X$. Deduce that $\mathbb{E}X < \infty$ if and only if

$$\sup_n \mathbb{E}(X \wedge n) < \infty.$$

- (b) [5 points] Show that if $\mathbb{E}X < \infty$, then $\mathbb{E}X \mathbb{1}_{\{X \geq n\}} \rightarrow 0$.

[Total for Question 4: 10 points]

5. [10 points] Recall that for a random variable/vector X in $\mathbb{R}^d$, its law is $\mu_X := P \circ X^{-1}$, that is, $\mu_X(A) = P(X \in A)$

Let X and Y be two independent random variables with laws μ and ν . Show that the law of the random vector (X, Y) is $\mu \times \nu$.

[Total for Question 5: 10 points]

6. Let X be a random variable taking value in $[0, \infty]$.

- (a) [5 points] Show that if $\mathbb{E}X < \infty$, then $P(X = \infty) = 0$.

Hint: use $\{X = \infty\} = \bigcap_{n=1}^{\infty} \{X \geq n\}$ and the inequality $\mathbb{E}X \geq \mathbb{E}X \mathbb{1}_{\{X \geq n\}} \geq nP(X \geq n)$.

- (b) [5 points] Show that if $\mathbb{E}X = 0$, then $X = 0$ almost surely, that is $P(X = 0) = 1$.

Hint: start from $\mathbb{E}X \geq \mathbb{E}X \mathbb{1}_{\{X \geq 1/n\}}$.

[Total for Question 6: 10 points]

7. Let $(\Omega_i, \mathcal{F}_i, P_i)$, $i = 1, 2$ be two probability spaces. Let $A \in \mathcal{F}_1 \otimes \mathcal{F}_2$. For $x \in \Omega_1$, we define the cross-section

$$A_x = \{y \in \Omega_2 : (x, y) \in A\}.$$

- (a) [5 points] Suppose that $(P_1 \times P_2)(A) = 0$. Show that $P_2(A_x) = 0$ for P_1 -almost every x .

Hint: apply Fubini to $f(x, y) = \mathbb{1}_A(x, y)$.

- (b) [5 points] Show that if $P_2(A_x) = 0$ for P_1 -almost every x , then $(P_1 \times P_2)(A) = 0$.

- (c) [5 points] Let μ, ν be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and let f be a Borel function. Define the graph of f by

$$\Gamma = \{(x, y) : y = f(x)\}.$$

Show that $\Gamma \in \mathcal{B}(\mathbb{R}^2)$ and if ν has no atoms, then

$$(\mu \times \nu)(\Gamma) = 0.$$

[Total for Question 7: 15 points]

8. (a) [5 points] Let $X_1, X_2, \dots$ be random variables taking values in $[0, \infty]$. Show that

$$\sum_{n=1}^{\infty} \mathbb{E}X_n = \mathbb{E} \sum_{n=1}^{\infty} X_n.$$

Hint: you can either use MCT, or use Fubini on $(\Omega \times \mathbb{N}, \mathcal{F} \otimes \mathcal{P}(\mathbb{N}), P \times \mu)$, where μ is the counting measure on $\mathbb{N}$.

(b) [5 points] Let $A_1, A_2, \dots$ be events and define an extended random variable

$$N(\omega) = \sum_{n=1}^{\infty} \mathbb{1}_{A_n}(\omega).$$

Show that if $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(N < \infty) = 1$.

[Total for Question 8: 10 points]